# A CASE STUDY REPORT ON

# Healthcare & Biology in the Cloud: Gene Sequencing, Protein Folding and ECG Analysis

**SUBMITTED TO THE SAVITRIBAI PHULE PUNE UNIVERSITY, PUNE**
**IN PARTIAL FULFILMENT OF THE REQUIREMENTS OF THE SUBJECT**

## CLOUD COMPUTING
## CASE STUDY

**THIRD YEAR AI & DS ENGINEERING**

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**Project Title: Healthcare & Biology in the Cloud: Gene Sequencing, Protein Folding and ECG Analysis**

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# ABSTRACT

Healthcare and biology now generate data at a scale that local computers struggle to handle, and cloud computing offers elastic, pay-per-use resources that suit such uneven demand. This report studies three representative workloads and how each is run in the cloud: gene sequencing, in which DNA reads are aligned to a reference and variants are called; protein folding, in which a one-dimensional amino-acid sequence is mapped to a three-dimensional structure using deep learning on GPU instances; and electrocardiogram (ECG) analysis, in which a continuous signal is converted into beats, heart rate and rhythm information.

The report proposes a cloud reference architecture, maps it to AWS, Microsoft Azure and OpenStack, and describes the pipeline, cloud design, advantages and limitations of each domain. Three small Python demonstrations were run locally on synthetic or toy data: global sequence alignment, exhaustive enumeration of a lattice protein model, and R-peak detection on a noisy synthetic ECG, including a noise-robustness test. These are illustrative and are not clinical or cloud benchmark results. The report concludes that the cloud makes all three workloads practical at scale, while privacy, cost control, generalisation and clinical validation remain the main challenges.

**Keywords:** *cloud computing, gene sequencing, NGS, protein folding, AlphaFold, ECG, Pan–Tompkins, AWS, healthcare analytics.*

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# 1. INTRODUCTION

## 1.1 Background

Modern medicine and biology produce data faster than local computers can analyse it. A human genome contains about three billion base pairs [1]; a protein of a few hundred residues can fold into an astronomically large number of shapes; and an ambulatory monitor records hundreds of thousands of heartbeats per day. Cloud computing, defined by NIST as on-demand, elastic, measured access to a shared pool of computing resources [2], fits this situation because the demand is bursty: a laboratory may need thousands of processor cores for a day and almost none for the next month [3], [4].

This report studies three representative workloads from Unit III (Healthcare & Biology) and asks how each is run on cloud platforms: gene sequencing (symbolic sequences), protein folding (three-dimensional structure) and ECG analysis (physiological time series).

## 1.2 Importance and Relevance

- **Precision medicine:** sequencing a patient's DNA reveals inherited disease risk and guides targeted therapy.
- **Drug discovery:** a protein's function depends on its 3D structure, so structure prediction supports target identification and enzyme design.
- **Cardiovascular care:** cardiovascular diseases are a leading cause of death worldwide [5], and the ECG is one of the cheapest diagnostic tests.
- **Access:** cloud-hosted analysis lets hospitals and laboratories without a data centre or in-house specialists use the same tools as large centres [6].

## 1.3 Scope of the Report

The report is conceptual and methodological. Section 4 to 7 explain the cloud architecture and the pipeline for each domain using standards, papers and provider documentation. Section 8 presents three small Python demonstrations that were run on a local computer on synthetic or toy data. **No cloud account was used and no cloud benchmark is claimed**; the cloud sections describe how such workloads are designed to run, not measured results. The report makes no clinical claims.

# 2. PROBLEM STATEMENT AND OBJECTIVES

**Problem.** Sequencers, structure predictors and wearable monitors generate data that is large, noisy and time-critical, and a fixed in-house cluster handles such uneven demand badly: it is either idle (wasted cost) or saturated (long queues). The three workloads also differ sharply in their resource profile, as Table 1 shows, so one design does not fit all.

**Table 1: Resource profile of the three workloads**

| Domain | Typical data | Compute profile | Cloud fit |
|---|---|---|---|
| Gene sequencing | Reads (FASTQ), tens to hundreds of GB per genome | CPU and memory heavy; independent per sample, so highly parallel | Batch fleets, object storage, workflow engines |
| Protein folding | One sequence, but multi-terabyte sequence databases for MSA search | I/O-bound database search, then GPU inference | Large-memory VMs, fast disks, GPU instances |
| ECG analysis | Continuous stream, small per patient but large in total | Light per beat; latency-sensitive for alarms | Stream ingestion, serverless functions, edge plus cloud |

**Objectives.** (i) Explain the biological background of the three domains; (ii) describe a cloud reference architecture and map it to AWS, Azure and OpenStack; (iii) describe the computational pipeline of each domain and how it uses cloud services; (iv) implement and run three small reproducible demonstrations; (v) discuss security, privacy, cost and limitations; and (vi) outline applications and future scope.

## 3. BIOLOGICAL BACKGROUND AND KEY CONCEPTS

**Central dogma.** DNA (alphabet A, C, G, T) is transcribed into RNA and translated, three bases (a codon) at a time, into a chain of amino acids [7]. Sequencing reads the DNA; protein folding describes what the resulting chain does next. **Folding.** Anfinsen showed that for many small proteins the sequence alone determines the native structure [8], while Levinthal pointed out that the number of possible conformations is far too large for random search [9]. **Cardiac signal.** Each heartbeat produces a P wave, a QRS complex and a T wave on the ECG; timing and shape changes reveal rhythm disorders.

- **Read and FASTQ:** a read is a short DNA fragment; FASTQ stores bases with a Phred quality score $Q = -10 \log_{10}(P)$, so Q30 means a 1-in-1000 error probability.
- **Alignment and variant:** alignment finds the best correspondence between two sequences; global alignment uses dynamic programming [10] and BLAST scales it by heuristics [11]. A variant is a difference from the reference, such as a SNP or small indel.
- **MSA and pLDDT:** a multiple sequence alignment of related proteins reveals residues that co-evolve and hence touch in 3D; pLDDT is AlphaFold's per-residue confidence score.
- **RR interval:** time between successive R peaks; heart rate = 60 / RR (seconds). MIT-BIH recordings are sampled at 360 Hz [12].

## 4. CLOUD PLATFORM FOUNDATION

### 4.1 Why the cloud suits these workloads

Four properties matter. **Elasticity:** hundreds of virtual machines can be started for a batch of genomes and released afterwards. **Pay-per-use:** cost follows consumption rather than peak capacity. **Scalable storage:** object storage holds petabytes with replication. **Specialised hardware:** GPU instances are available without capital purchase. Public datasets (for example the AlphaFold database [13]) can also be hosted next to the compute, which avoids repeated downloads.

### 4.2 Reference architecture

Figure 1 shows the architecture used throughout this report. Data lands in object storage; compute is started on demand (CPU batch fleets for alignment, GPU nodes for structure prediction and neural networks); managed services (queues, stream processing, workflow engines) connect the stages; and results go to a database or dashboard. Identity management, encryption and audit logging apply to every layer.

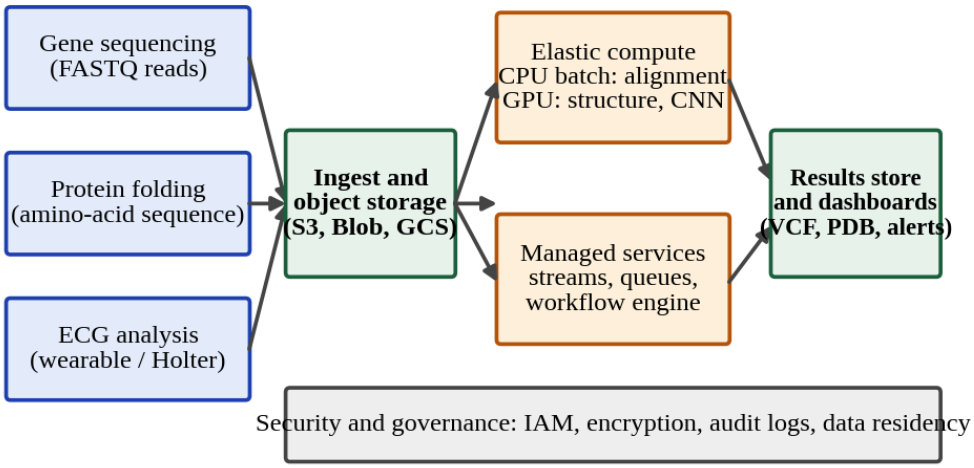



**Figure 1: Cloud reference architecture for healthcare and biology workloads**

### 4.3 Mapping to platforms

**Table 2: Workload needs mapped to platform services**

| Need | AWS | Microsoft Azure | OpenStack (private) |
|---|---|---|---|
| Raw data storage | S3 | Blob Storage | Swift |
| Batch virtual machines | EC2, AWS Batch | Virtual Machines, Azure Batch | Nova |
| GPU compute | EC2 GPU instance types | NC / ND-series VMs | Nova with GPU passthrough |
| Metadata and results | RDS, DynamoDB (SimpleDB is legacy) | SQL Azure (Azure SQL Database) | Trove |
| Streaming ingest | Kinesis, IoT Core | Event Hubs, IoT Hub | Third-party broker (e.g. Kafka) |
| Identity and access | IAM | Microsoft Entra ID | Keystone |

OpenStack, CloudStack and Eucalyptus matter when a hospital must keep patient data on premises: a private cloud gives the same self-service model while data stays under the institution's control [14]. Hybrid designs keep identifiable data private and send de-identified data to a public cloud for heavy computation.

## 5. GENE SEQUENCING IN THE CLOUD

**Pipeline.** In short-read sequencing, DNA is fragmented, adapters are added and fragments are read by synthesis on a flow cell, producing FASTQ files [15], [16]. The reads are quality-checked, aligned to a reference genome with a Burrows–Wheeler aligner such as BWA [17], and variants are called with statistical frameworks such as GATK [18] or deep-learning callers such as DeepVariant [19] (Figure 2).

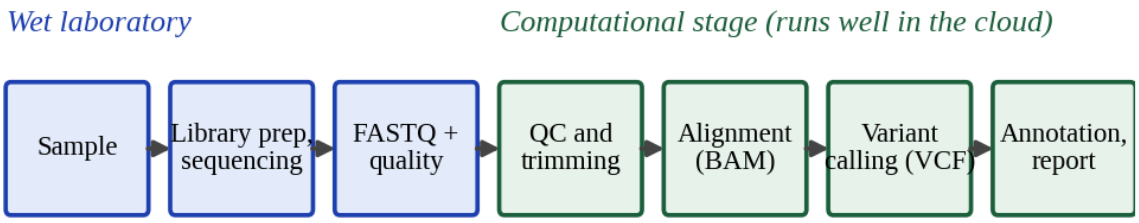



**Figure 2: Typical NGS workflow; the computational stage is the part moved to the cloud**

**Why it parallelises well.** Reads are independent, so alignment can be split across many machines; CloudBurst did this with MapReduce on Amazon EC2 [20]. In practice a workflow engine (for example Nextflow or WDL) launches one job per sample or genomic region on a batch fleet. Because jobs are restartable, they can run on interruptible Spot capacity at a lower price [21]. Managed services such as AWS HealthOmics package storage and workflow execution for omics data [22]. Raw FASTQ files, which are rarely re-read, can move to cheaper archive storage after alignment, while alignments and variants stay on standard storage.

**Advantages:** fast turnaround for large cohorts, no idle cluster, easy data sharing between collaborators [6]. **Limitations:** genomes are identifying personal data (Section 9), transfer of hundreds of gigabytes per genome takes time and money, and long-running pipelines need careful cost control.

## 6. PROTEIN FOLDING IN THE CLOUD

Protein structure prediction infers the 3D shape of a chain from its sequence. Experimental methods are accurate but slow and costly, and deep learning has changed the field, as Table 3 shows. AlphaFold 2 reached accuracy competitive with experiments in the CASP14 assessment [23], and RoseTTAFold uses a related three-track network [24].

**Table 3: Approaches to protein structure determination**

| Method | Basis | Strength | Limitation |
|---|---|---|---|
| X-ray crystallography | Diffraction from crystals | Atomic resolution | Needs crystals; static |
| Cryo-EM | Electron imaging of frozen particles | Large complexes | Costly equipment |
| Homology modelling | Template from related protein | Fast with a template | Fails without homologues |
| Deep learning (AlphaFold, RoseTTAFold) | Learned from known structures and MSAs | High accuracy for many chains; minutes to hours | Weak for dynamics, ligands, novel folds |

**Cloud design.** AlphaFold 2 first searches large sequence databases to build an MSA, then a transformer-like network (the Evoformer) exchanges information between the MSA and residue-pair representations, a structure module outputs atomic coordinates, and the result is recycled several times [23]. The two stages have opposite needs: the database search is memory- and disk-intensive and runs on large CPU instances with fast storage, while neural inference needs GPU instances. Separating them lets each use the cheapest suitable instance type. ColabFold replaces the slow search with a faster method, reducing cost [25], and

precomputed structures for many proteins are already hosted publicly [13], so a user often needs to download rather than compute.

**Limitations:** a predicted structure is a static model; low-confidence regions (low pLDDT) must not be over-interpreted; and the multi-terabyte databases make private setup expensive, which is another reason to use hosted services.

## 7. ECG ANALYSIS IN THE CLOUD

**Algorithm.** The classic Pan–Tompkins detector [26] band-pass filters the ECG (about 5–15 Hz) to emphasise the QRS complex, differentiates it, squares it, smooths it with a moving-window integrator and applies adaptive thresholds; R-peak positions give RR intervals, heart rate and variability measures such as SDNN. Beyond rules, machine-learning classifiers use hand-crafted features, and a deep network trained on a large single-lead dataset has been reported to classify rhythms at a level comparable to cardiologists on its test set [27]. Public data such as MIT-BIH [12] and PhysioNet [28] support development.

**Cloud design.** A wearable sends samples to a phone, which forwards them to a cloud ingestion service; stream processing filters and detects beats; an alarm service notifies clinicians; and raw data is archived. A sensible split is **edge for latency, cloud for scale**: simple detection and alarms run on the device or phone, while cloud services train models on pooled data, run heavier classifiers, and keep long-term records. The volume is easy to underestimate, as Table 6 shows.

**Limitations:** sensitivity to motion noise, class imbalance, poor generalisation across devices and populations, limited interpretability of deep models, and the need for regulatory approval before any alert is used clinically.

## 8. IMPLEMENTATION AND RESULTS

The demonstrations use Python 3 with NumPy, SciPy and Matplotlib and were run locally. They are small on purpose so that each step can be inspected. **They are not clinical results and not benchmarks of the real systems in Sections 5 to 7.**

### 8.1 Sequence alignment

A Needleman–Wunsch aligner [10] (match +1, mismatch −1, gap −2) aligned the reference ATGGCGTACGTTAGC with the read ATGGCTACGTTTAGC. The core of the algorithm is shown below.

```
def needleman_wunsch(a, b, match=1, mismatch=-1, gap=-2):
    n, m = len(a), len(b)
    S = [[0]*(m+1) for _ in range(n+1)]
    for i in range(1, n+1): S[i][0] = i*gap
    for j in range(1, m+1): S[0][j] = j*gap
    for i in range(1, n+1):
        for j in range(1, m+1):
            diag = S[i-1][j-1] + (match if a[i-1]==b[j-1] else mismatch)
            S[i][j] = max(diag, S[i-1][j] + gap, S[i][j-1] + gap)
    return S   # traceback from S[n][m] gives the alignment
```

*Listing 1: Dynamic-programming core of the aligner*

```
Ref : ATGGCGTACG-TTAGC
      ||||| |||| |||||
Read: ATGGC-TACGTTTAGC      (score = 10)
```

*Output 1: Alignment returned by the program*

Of 16 columns, 14 are matches and two are gaps: the read lacks one G near position 6 and has an extra T near position 11, which looks like a pair of small indels. In a real pipeline such a pattern would be checked against base quality and neighbouring reads before being reported as a variant.

## 8.2 Lattice protein model

In the hydrophobic–polar (HP) model [29] a chain lies on a 2D square lattice as a self-avoiding walk, and each pair of non-adjacent H residues that touch scores −1. Depth-first search enumerated all walks with the first step fixed. The count grows by about 2.7 times per added residue (Figure 3 a): 1,604,149 conformations at 16 residues versus 593,611 at 15, and extrapolation gives about $10^{41}$ to $10^{42}$ for 100 residues, so exhaustive search is impossible. This is the combinatorial core of Levinthal's argument [9]. For the 12-residue sequence HPHPPHHPHPPH all 30,073 conformations were scored; the lowest energy was −5, reached by 4 conformations (Figure 3 b).

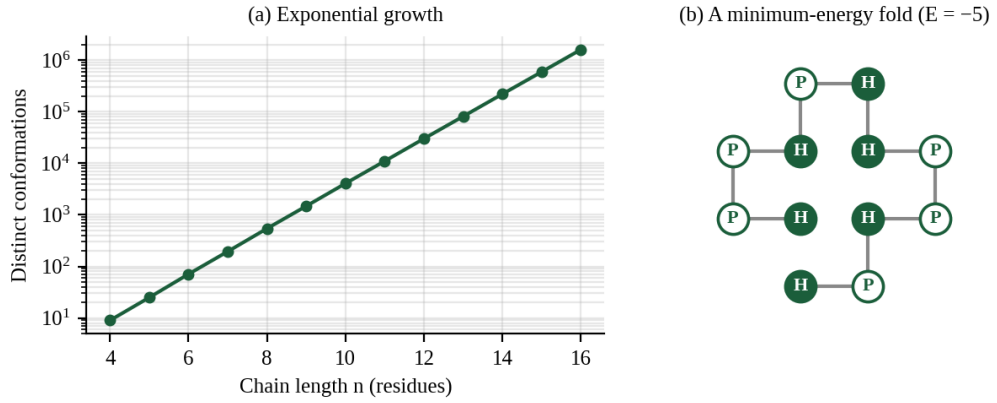



**Figure 3: Lattice protein: (a) growth of conformations with chain length; (b) a minimum-energy fold**

## 8.3 ECG R-peak detection

A ten-second synthetic ECG at 360 Hz was generated from Gaussian P, Q, R, S and T components with slightly varying beat intervals; baseline wander (0.3 Hz), 50 Hz mains interference and Gaussian noise were added. The detector follows Section 7 (band-pass 5–15 Hz, derivative, squaring, 150 ms moving-window integration, thresholding at 35% of the maximum, 250 ms refractory distance). Figure 4 shows the stages and Table 4 the outcome at the base noise level.

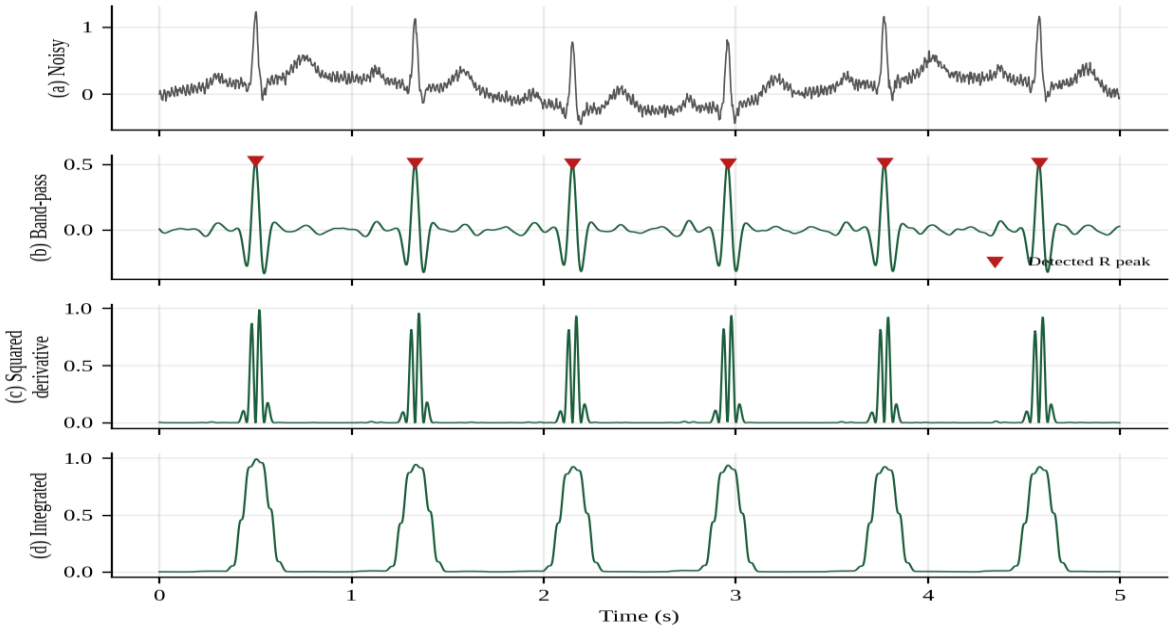



**Figure 4: ECG stages on the first 5 s: (a) noisy input, (b) band-pass with detected R peaks, (c) squared derivative, (d) integrated signal**

**Table 4: Detection results (10 s synthetic ECG, 360 Hz, 50 ms tolerance)**

| Quantity | Value |
|---|---|
| True beats / detected beats | 12 / 12 |
| True positives / false negatives / false positives | 12 / 0 / 0 |
| Sensitivity and precision | 100% and 100% |
| Mean heart rate (detected / generated) | 72.9 / 72.9 beats/min |
| SDNN (detected / generated) | 13.5 ms / 13.7 ms |

A perfect score on one easy signal proves little, so the same detector was re-run with stronger Gaussian noise (R-wave amplitude = 1.0). Table 5 shows that it holds up to moderate noise and then degrades, first by missing beats and then by adding false detections. Real recordings add ectopic beats, varying morphology and motion artefacts, so performance on MIT-BIH would be lower still.

**Table 5: Effect of noise level on detection (same 12 true beats)**

| Noise std. dev. | Detected | True pos. | False pos. | False neg. |
|---|---|---|---|---|
| 0.1 | 12 | 12 | 0 | 0 |
| 0.3 | 12 | 12 | 0 | 0 |
| 0.6 | 9 | 9 | 0 | 3 |
| 1 | 11 | 6 | 5 | 6 |

### 8.4 What the demonstrations show

The aligner exposes small insertions and deletions; the lattice count shows why brute-force folding fails; and the filter-and-threshold detector works on clean signals and fails gracefully as noise rises. Cloud relevance is indirect: all three are small versions of workloads that become practical at scale only with parallel resources, Table 6 gives illustrative data-volume arithmetic (assumptions stated, not measurements).

**Table 6: Illustrative data-volume arithmetic (assumptions in the third column)**

| Workload | Result | Assumption |
|---|---|---|
| ECG, one patient | about 43 MB per day | 1 lead, 250 Hz, 2 bytes per sample |
| ECG, 10,000 patients | about 430 GB per day, about 158 TB per year | Same rate, continuous recording |
| Genomes, 1,000 patients | about 100 TB of raw reads | About 100 GB per 30× genome (order of magnitude) |

## 9. SECURITY, PRIVACY, ADVANTAGES AND LIMITATIONS

Genomes and ECGs are personal health data, and a genome cannot be changed if leaked. Under the shared-responsibility model the provider secures the infrastructure while the customer must configure identity, encryption and access. Minimum controls are strong authentication and least-privilege roles, encryption at rest and in transit with customer-managed keys, audit logging, de-identification before sharing, and data residency within India where required. India's Digital Personal Data Protection Act, 2023 requires lawful processing and consent for personal data [30]. Cloud platforms also bring vendor lock-in and data-egress charges.

**Table 7: Advantages and limitations of cloud-based healthcare analysis**

| Advantages | Limitations |
|---|---|
| Elastic capacity for bursty workloads | Privacy, consent and residency obligations |
| No capital cost; pay per use | Costs can escalate without monitoring; egress fees |
| Access to GPUs and hosted public datasets | Large transfers are slow; lock-in to provider tools |
| Easy collaboration and reproducible workflows | Models need clinical validation before use |
| Built-in redundancy and audit logging | Connectivity dependence; alarms need edge fallback |

## 10. APPLICATIONS AND FUTURE SCOPE

- **Applications:** cloud genomics for rare-disease diagnosis and tumour profiling; structure-guided drug and enzyme design; remote cardiac monitoring; pathogen genomic surveillance.
- **Future scope:** combining genomic, structural and physiological data to predict how a variant affects a patient; modelling protein dynamics and complexes; self-supervised models that reduce the need for labelled data; privacy-preserving federated learning across hospitals; and trustworthy AI with uncertainty estimates and prospective clinical trials.

## 11. CONCLUSION

Gene sequencing, protein folding and ECG analysis share one pattern: raw, high-volume data is cleaned, modelled and validated. They differ in resource profile (parallel CPU batches, database search plus GPU inference, and low-latency streams), and the cloud serves each through elastic compute, object storage and managed services. The local demonstrations confirm three basic points: dynamic programming exposes small indels, conformation counts grow exponentially, and a classical detector works on clean ECG and degrades under heavy noise. The practical limits are privacy, cost control, generalisation and clinical validation rather than raw computing power.

## 12. REFERENCES

[1] E. S. Lander et al., “Initial sequencing and analysis of the human genome,” Nature, vol. 409, pp. 860–921, 2001.

[2] P. Mell and T. Grance, “The NIST definition of cloud computing,” NIST Special Publication 800-145, Sep. 2011.

[3] M. Armbrust et al., “A view of cloud computing,” Commun. ACM, vol. 53, no. 4, pp. 50–58, 2010.

[4] L. D. Stein, “The case for cloud computing in genome informatics,” Genome Biol., vol. 11, art. 207, 2010.

[5] World Health Organization, “Cardiovascular diseases (CVDs),” fact sheet. [Online]. Available: https://www.who.int/news-room/fact-sheets/detail/cardiovascular-diseases-(cvds) (accessed [[ENTER DATE]]).

[6] B. Langmead and A. Nellore, “Cloud computing for genomic data analysis and collaboration,” Nat. Rev. Genet., vol. 19, pp. 208–219, 2018.

[7] B. Alberts et al., Molecular Biology of the Cell, 6th ed. New York, NY, USA: Garland Science, 2014.

[8] C. B. Anfinsen, “Principles that govern the folding of protein chains,” Science, vol. 181, no. 4096, pp. 223–230, 1973.

[9] C. Levinthal, “Are there pathways for protein folding?” J. Chim. Phys., vol. 65, pp. 44–45, 1968.

[10] S. B. Needleman and C. D. Wunsch, “A general method applicable to the search for similarities in the amino acid sequence of two proteins,” J. Mol. Biol., vol. 48, no. 3, pp. 443–453, 1970.

[11] S. F. Altschul, W. Gish, W. Miller, E. W. Myers, and D. J. Lipman, “Basic local alignment search tool,” J. Mol. Biol., vol. 215, no. 3, pp. 403–410, 1990.

[12] G. B. Moody and R. G. Mark, “The impact of the MIT-BIH Arrhythmia Database,” IEEE Eng. Med. Biol. Mag., vol. 20, no. 3, pp. 45–50, 2001.

[13] M. Varadi et al., “AlphaFold Protein Structure Database: massively expanding the structural coverage of protein-sequence space with high-accuracy models,” Nucleic Acids Res., vol. 50, no. D1, pp. D439–D444, 2022.

[14] OpenStack Foundation, “OpenStack documentation.” [Online]. Available: https://docs.openstack.org/

[15] S. Goodwin, J. D. McPherson, and W. R. McCombie, “Coming of age: ten years of next-generation sequencing technologies,” Nat. Rev. Genet., vol. 17, pp. 333–351, 2016.

[16] S. Behjati and P. S. Tarpey, “What is next generation sequencing?” Arch. Dis. Child. Educ. Pract. Ed., vol. 98, no. 6, pp. 236–238, 2013.

[17] H. Li and R. Durbin, “Fast and accurate short read alignment with Burrows–Wheeler transform,” Bioinformatics, vol. 25, no. 14, pp. 1754–1760, 2009.

[18] G. A. Van der Auwera et al., “From FastQ data to high-confidence variant calls: the Genome Analysis Toolkit best practices pipeline,” Curr. Protoc. Bioinformatics, vol. 43, pp. 11.10.1–11.10.33, 2013.

[19] R. Poplin et al., “A universal SNP and small-indel variant caller using deep neural networks,” Nat. Biotechnol., vol. 36, pp. 983–987, 2018.

[20] M. C. Schatz, “CloudBurst: highly sensitive read mapping with MapReduce,” Bioinformatics, vol. 25, no. 11, pp. 1363–1369, 2009.

[21] Amazon Web Services, “Amazon EC2 User Guide” (instance types and Spot Instances). [Online]. Available: https://docs.aws.amazon.com/ec2/

[22] Amazon Web Services, “AWS HealthOmics User Guide.” [Online]. Available: https://docs.aws.amazon.com/omics/

[23] J. Jumper et al., “Highly accurate protein structure prediction with AlphaFold,” Nature, vol. 596, pp. 583–589, 2021.

[24] M. Baek et al., “Accurate prediction of protein structures and interactions using a three-track neural network,” Science, vol. 373, no. 6557, pp. 871–876, 2021.

[25] M. Mirdita et al., “ColabFold: making protein folding accessible to all,” Nat. Methods, vol. 19, pp. 679–682, 2022.

[26] J. Pan and W. J. Tompkins, “A real-time QRS detection algorithm,” IEEE Trans. Biomed. Eng., vol. BME-32, no. 3, pp. 230–236, Mar. 1985.

[27] A. Y. Hannun et al., “Cardiologist-level arrhythmia detection and classification in ambulatory electrocardiograms using a deep neural network,” Nat. Med., vol. 25, pp. 65–69, 2019.

[28] A. L. Goldberger et al., “PhysioBank, PhysioToolkit, and PhysioNet: components of a new research resource for complex physiologic signals,” Circulation, vol. 101, no. 23, pp. e215–e220, 2000.

[29] K. F. Lau and K. A. Dill, “A lattice statistical mechanics model of the conformational and sequence spaces of proteins,” Macromolecules, vol. 22, no. 10, pp. 3986–3997, 1989.

[30] Government of India, “The Digital Personal Data Protection Act, 2023,” Act No. 22 of 2023.